Axisymmetric Cartoon Z4c in AthenaK: Implementation, Qualification, and Current-Gauge Analogues of arXiv:2607.10843v1

Reproducibility evidence report

Evidence report updated: 2026-08-13

Abstract

This modular document records the verified source and process evidence for the axisymmetric Cartoon Z4c report. The authoritative pushed source is commit `5760da5893b256d32da5fb13f92ed8e0c102ed39`, tree `9318e083497997edb80d93c913934acd987699eb`. R1 is accepted and R2 is accepted with an explicit lower-basis waiver, while its original all-basis gate is not passed. Brill Figure 3 passed import but is unresolved after a late evolved-state failure at N128 cycle 544; N192/N256 did not run. The authoritative time-asymmetric V2 pool is blocked by its unchanged ADM-stability gate. Final R5 v19 closes the preceding CUDA out-of-bounds defect but fails closed on 200 invalid chi parent stencils after the first RK stage; candidate timing never began and R5 is not accepted. Milestone 2 is waived, not passed, and R3–R5 remain incomplete or blocked. A later half-plane Kerr experiment used the requested pre-collapsed lapse with default 1+log and direct Gamma-driver shift: $M/32$ reached $5M$, but $M/48$ failed at $4.5604M$ in a diagnosed puncture-interior Gamma/A–Ricci mode, so $M/64$ was not run and no convergence qualification follows. Every future comparison to arXiv:2607.10843v1 must be labeled an AthenaK current-gauge analogue and backed by immutable machine-readable data.

Contents

1 Executive summary and claim boundary

The accepted prerequisite is an interpolation-free signed-radius SO(2) derivative provider and a 171-operator manufactured oracle. Its exact inventory is 137 truncating, 12 exact identity, 12 exact plane-algebraic, and 10 exact-discrete operators. No production derivative-formula defect was demonstrated. The separate derivative report and its failed-qualification records remain immutable.

The authoritative AthenaK source is commit `5760da5893b256d32da5fb13f92ed8e0c102ed39`, tree `9318e083497997edb80d93c913934acd987699eb`. It contains the logical R1/R2 implementation slices, campaign tooling, the reusable Kerr puncture pgen, and the Cartoon $m = 0$ FastFlow adapter with restart schema 2. Focused static, mathematical, and self-tests passed locally. The current remote sequence is deliberately retained rather than collapsed into a single success/failure label:

- v5, job 56666406, failed during FastFlow CUDA compilation after configuration; no executable, focused test, or numerical case ran;
- v6, job 56668445, completed the full CUDA build and verified the CUDA+MPI executable backend, then failed in the seven-test focused harness; three Python tests were dispatched through CMake's legacy Python-3.6 interpreter and one preallocation fixture expected the superseded diagnostic ordering. No R1 or R2 numerical case ran;
- commits `ca5ede9b0a357380218d21cb29362e773ac914c6` and `138f4d54e20bfb70b9d2691b82b022e4de06d956` contain the narrow test-only fixture repairs. The v7 launch deck also pins CMake's legacy `PYTHON_EXECUTABLE` to the checksum-bound Python 3.11 executable;
- later campaigns closed R1 and supplied the user-authorized R2 disposition "accepted with an explicit lower-basis waiver." The original all-basis gate is not passed: displaced $(4, 16)$ remains failed evidence, while displaced $\pm(6, 24)$ and its mirror relation are the required accepted off-axis evidence. The L8 supplement is corroborating characterization.
- R5 v10, job 56726255, passed the three current-source linear functional cases and produced a stable five-sample baseline characterization, then failed `boosted_n128` at cycle 1 before candidate timing. The strict regrid gate found 108 invalid chi parent stencils. The diagnosis identified the ordinary coarse/fine boundary path, which had applied raw high-order interpolation to chi. The authoritative source routes only chi through the existing whole-sibling positive fallback and retains fail-closed parent checks, without flooring, clipping, or threshold changes.
- R5 v18, job 56741854, exposed a CUDA out-of-bounds access in the newly re-enabled same-level coarse restriction. The complete-fine-pair and oriented outer-edge repair is integrated at the authoritative tip. R5 v19, job 56748297, verified that this CUDA defect is closed: initialization completed with 194,096 high-order groups, 16 positive limited groups, and no invalid parents. The first post-RK boundary pass then failed closed on 200 invalid parent stencils.

The evidence cannot distinguish invalid evolved chi from its coarse restriction representation, proves no further single copy/order seam, and does not support a physical-instability claim. No candidate timing ran and R5 is not accepted.

- A later independent half-plane Kerr resolution experiment used pushed source `f95a05802621b76cff2894d562c38df4b0d09661`, pre-collapsed lapse, default advective 1+log, the direct Gamma-driver shift, and `dchi_max=0.02`. The $M/32$ grid reached $5M$, but the $M/48$ grid failed at $4.5604M$ in a reproducible puncture-interior Gamma/A–Ricci mode. The $M/64$ grid was not run after convergence had already failed. This is diagnosed failure evidence, not a new R2 pass or a three-grid convergence result.

None of these build or process facts is converted into a runtime result.

The user waived the original common-integration Milestone 2 as a standalone completion gate *after accepted implementation and characterization, without full convergence or backend qualification*. This report does not rename that waiver as a pass. In particular, it does not imply that Cartoon evolution, AMR, restart, horizons, CPU rank consistency, or CUDA execution has already been qualified.

Table 1: Status at the final evidence freeze. Failed or blocked gates are retained and do not support numerical qualification claims.

Gate	Status	Allowed statement
Derivative implementation	accepted prerequisite	implementation only
Milestone 2	waived by user	explicitly not passed
R1 AMR/restart/diagnostics	accepted campaign evidence	source-bound CUDA+MPI result
R2 Kerr puncture/horizon	accepted with lower-basis waiver	all-basis gate not passed
Half-plane Kerr resolution	$M/48$ fails before $5M$	no convergence qualification
R3 Brill Figure 3	N128 late-state failure	unresolved; no accepted curve
R3 time-asymmetric V2	candidate pool exhausted; blocked	no promoted payload or fate
R4 literature freeze	verified	source/amplitude provenance only
R4 Figures 1/3/4/5	blocked or unresolved	no accepted AthenaK analogue
R5 regression/performance	v19 terminal limitation	not accepted; no performance verdict
Report/reproducibility product	final evidence freeze	limitations retained

The Brill target comparisons preserve their declared AthenaK gauge rather than tuning it after observing a result. The separate half-plane Kerr resolution experiment uses the user-requested default 1+log/direct Gamma-driver gauge and is labeled accordingly. Every generated paper plot and quantitative comparison will carry its exact gauge provenance and the label "AthenaK current-gauge analogue." Coordinate-time agreement with bamps, prague, or sphGR will not be claimed.

2 Literature and mathematical formulation

2.1 From Cartoon to analytic dimensional reduction

The original Cartoon construction evolves a Cartesian slab only one finite-difference molecule thick in the suppressed direction. Axisymmetry fills the off-plane points by rotating data from the meridional plane, with a one-dimensional interpolation in cylindrical radius, after which ordinary Cartesian stencils are applied [2]. This avoids a cylindrical coordinate basis at the axis, but interpolation remains part of the derivative operator. Pretorius instead evaluates suppressed derivatives analytically from the rotational Killing equation, eliminating the thin-slab interpolation and making the reduction easier to combine with adaptive mesh refinement [6]. The modified-Cartoon literature subsequently tabulated the general scalar, vector, and tensor identities and, critically, the regularized axis limits [3].

Modern axisymmetric relativity codes demonstrate that Cartesian Cartoon can be combined with sixth-order differencing and Z4c constraint propagation. For example, SACRA-2D uses an interpolated slab and gives analytic mixed-derivative relations at refinement edges [4]. This is useful independent sign evidence, but it is not the architecture adopted here: the production backend is interpolation-free throughout. The bamps calculations in the target comparison paper likewise use the Pretorius analytic variant [1].

2.2 Coordinates and Killing identities

We use Cartesian coordinates

$$(x^1, x^2, x^3) = (x, z, y) = (\rho_{\text{signed}}, z_{\text{phys}}, y_{\text{suppressed}}) \tag{1}$$

on the full meridional plane $y = 0$. The physical cylindrical radius is $R = (x^2 + y^2)^{1/2}$, so $R = |x|$ on the plane. Rotations about the physical z axis are generated by

$$\xi = -y\partial_x + x\partial_y. \tag{2}$$

For an invariant field $\mathcal{L}_\xi T = 0$. At $y = 0$ and nonzero signed $\rho = x$, define $J(V_x, V_z, V_y) = (-V_y, 0, V_x)$. Then

$$\partial_y f = 0, \qquad \partial_y V = \frac{JV}{\rho}, \qquad \partial_y T = \frac{JT + TJ^{\mathsf{T}}}{\rho}. \tag{3}$$

The same Cartesian component equations hold for all-upper or all-lower indices because the rotation is orthogonal. A genuinely mixed-index tensor must receive the generator action appropriate to each index.

For a scalar,

$$f_{,yy} = \frac{f_{,x}}{\rho}, \qquad f_{,xy} = f_{,zy} = 0. \tag{4}$$

For the compact component tables below, suffixes in V_x, V_z, V_y are Cartesian component *labels*, not covariant index positions. The table applies component-by-component to either a contravariant vector or a covector under rigid Cartesian rotation. Metric raising and lowering is a separate operation. For $V = (V_x, V_z, V_y)$, the nonzero suppressed first derivatives are

$$V_{x,y} = -\frac{V_y}{\rho}, \qquad V_{z,y} = 0, \qquad V_{y,y} = \frac{V_x}{\rho}, \tag{5}$$

and the suppressed second derivatives are

$$V_{x,yy} = \frac{V_{x,x}}{\rho} - \frac{V_x}{\rho^2}, \qquad V_{z,yy} = \frac{V_{z,x}}{\rho}, \qquad V_{y,yy} = \frac{V_{y,x}}{\rho} - \frac{V_y}{\rho^2}. \tag{6}$$

The radial–suppressed mixed derivatives are

$$\begin{aligned} V_{x,xy} &= -\frac{V_{y,x}}{\rho} + \frac{V_y}{\rho^2}, & V_{z,xy} &= 0, & V_{y,xy} &= \frac{V_{x,x}}{\rho} - \frac{V_x}{\rho^2}, \\ V_{x,zy} &= -\frac{V_{y,z}}{\rho}, & V_{z,zy} &= 0, & V_{y,zy} &= \frac{V_{x,z}}{\rho}. \end{aligned} \tag{7}$$

Consequently

$$\partial_i V^i = \partial_x V^x + \partial_z V^z + \frac{V^x}{\rho}. \tag{8}$$

This divergence equation does not identify V^i with a metric-lowered V_i.

For a symmetric tensor, the first suppressed derivatives are

$$\begin{aligned} T_{xx,y} &= -\frac{2T_{xy}}{\rho}, & T_{xy,y} &= \frac{T_{xx} - T_{yy}}{\rho}, & T_{yy,y} &= \frac{2T_{xy}}{\rho}, \\ T_{xz,y} &= -\frac{T_{yz}}{\rho}, & T_{yz,y} &= \frac{T_{xz}}{\rho}, & T_{zz,y} &= 0. \end{aligned} \tag{9}$$

The second suppressed derivatives are

$$\begin{aligned} T_{xx,yy} &= \frac{T_{xx,x}}{\rho} - \frac{2(T_{xx} - T_{yy})}{\rho^2}, & T_{xy,yy} &= \frac{T_{xy,x}}{\rho} - \frac{4T_{xy}}{\rho^2}, \\ T_{yy,yy} &= \frac{T_{yy,x}}{\rho} + \frac{2(T_{xx} - T_{yy})}{\rho^2}, & T_{xz,yy} &= \frac{T_{xz,x}}{\rho} - \frac{T_{xz}}{\rho^2}, \\ T_{yz,yy} &= \frac{T_{yz,x}}{\rho} - \frac{T_{yz}}{\rho^2}, & T_{zz,yy} &= \frac{T_{zz,x}}{\rho}. \end{aligned} \tag{10}$$

Radial–suppressed mixed derivatives are

$$T_{xx,xy} = -\frac{2T_{xy,x}}{\rho} + \frac{2T_{xy}}{\rho^2}, \tag{11}$$

$$\begin{aligned} T_{xy,xy} &= \frac{T_{xx,x} - T_{yy,x}}{\rho} - \frac{T_{xx} - T_{yy}}{\rho^2}, \\ T_{yy,xy} &= \frac{2T_{xy,x}}{\rho} - \frac{2T_{xy}}{\rho^2}, \end{aligned} \tag{12}$$

$$T_{xz,xy} = -\frac{T_{yz,x}}{\rho} + \frac{T_{yz}}{\rho^2}, \qquad T_{yz,xy} = \frac{T_{xz,x}}{\rho} - \frac{T_{xz}}{\rho^2}, \qquad T_{zz,xy} = 0. \tag{13}$$

Axial–suppressed derivatives contain no derivative of $1/\rho$:

$$\begin{aligned} T_{xx,zy} &= -\frac{2T_{xy,z}}{\rho}, & T_{xy,zy} &= \frac{T_{xx,z} - T_{yy,z}}{\rho}, & T_{yy,zy} &= \frac{2T_{xy,z}}{\rho}, \\ T_{xz,zy} &= -\frac{T_{yz,z}}{\rho}, & T_{yz,zy} &= \frac{T_{xz,z}}{\rho}, & T_{zz,zy} &= 0. \end{aligned} \tag{14}$$

Advective derivatives retain the suppressed component:

$$\beta^i \partial_i U = \beta^x U_{,x} + \beta^z U_{,z} + \beta^y U_{,y}, \tag{15}$$

where $U_{,y}$ is evaluated with the scalar-, vector-, or tensor-aware identity. The fact that no y grid is stored does not set tensor component derivatives to zero.

2.3 Signed-radius parity and regular axis

A rotation by π maps ρ to $-\rho$. Scalars, V_z, and $T_{xx}, T_{xy}, T_{yy}, T_{zz}$ are even; V_x, V_y, T_{xz}, T_{yz} are odd. Smoothness strengthens this to

$$V_x, V_y, T_{xz}, T_{yz} = O(\rho), \quad T_{xy} = O(\rho^2), \quad T_{xx} - T_{yy} = O(\rho^2). \tag{16}$$

Twist-free reflection also sets $V_y = T_{xy} = T_{yz} = 0$ on the meridional plane, but that is a property of the paper's data, not of the reusable Z4c backend.

Taking two-sided limits yields the complete scalar table

$$f_{,y} = 0, \qquad f_{,yy} = f_{,xx}, \qquad f_{,xy} = 0, \qquad f_{,zy} = 0. \tag{17}$$

For vectors,

$$\begin{aligned}
V_{x,y} &= -V_{y,x}, & V_{z,y} &= 0, & V_{y,y} &= V_{x,x}, \\
V_{x,yy} &= 0, & V_{z,yy} &= V_{z,xx}, & V_{y,yy} &= 0, \\
V_{x,xy} &= 0, & V_{z,xy} &= 0, & V_{y,xy} &= 0, \\
V_{x,zy} &= -V_{y,xz}, & V_{z,zy} &= 0, & V_{y,zy} &= V_{x,xz}.
\end{aligned} \tag{18}$$

The axis divergence is $\partial_i V^i = 2\partial_x V^x + \partial_z V^z$. For symmetric tensors, the first-suppressed limits are

$$\begin{aligned}
T_{xx,y} &= 0, & T_{xy,y} &= 0, & T_{yy,y} &= 0, & T_{zz,y} &= 0, \\
T_{xz,y} &= -T_{yz,x}, & T_{yz,y} &= T_{xz,x}.
\end{aligned} \tag{19}$$

The second-suppressed limits are

$$\begin{aligned}
T_{xx,yy} &= T_{yy,xx}, & T_{yy,yy} &= T_{xx,xx}, & T_{xy,yy} &= -T_{xy,xx}, & T_{zz,yy} &= T_{zz,xx}, \\
T_{xz,yy} &= 0, & T_{yz,yy} &= 0.
\end{aligned} \tag{20}$$

The mixed limits are

$$\begin{aligned}
T_{xx,xy} &= -T_{xy,xx}, & T_{xy,xy} &= \tfrac{1}{2}(T_{xx,xx} - T_{yy,xx}), & T_{yy,xy} &= T_{xy,xx}, \\
T_{xz,xy} &= 0, & T_{yz,xy} &= 0, & T_{zz,xy} &= 0, \\
T_{xx,zy} &= 0, & T_{xy,zy} &= 0, & T_{yy,zy} &= 0, & T_{zz,zy} &= 0, \\
T_{xz,zy} &= -T_{yz,xz}, & T_{yz,zy} &= T_{xz,xz}.
\end{aligned} \tag{21}$$

Composite $1/\rho$ and $1/\rho^2$ expressions must be replaced by these limits before floating-point evaluation; separately evaluating singular terms and subtracting them loses the regular cancellation [3].

2.4 Independent analytic oracle

The archived oracle uses $s = x^2 + y^2$, $R = (x, y)$, and $P = (-y, x)$ and constructs

$$f = F(s, z), \qquad V_\perp = a(s, z)R + b(s, z)P, \qquad V_z = c(s, z), \tag{22}$$

$$\begin{aligned}
T = {}& pI_\perp + qe_ze_z + r(Re_z + e_zR) + u(Pe_z + e_zP) \\
& + v(RR - PP) + w(RP + PR),
\end{aligned} \tag{23}$$

with independent high-degree polynomial coefficients, including high powers of z and mixed s–z terms. Exact expectations come from formal polynomial differentiation in a standalone script with no AthenaK imports. For every scalar, vector component, tensor component, and each of $\partial_x, \partial_z, \partial_{xx}, \partial_{zz}, \partial_{xz}$ at both $\rho = \pm 0.73$, the final observed orders lie in $[1.824, 2.034]$, $[3.992, 4.013]$, and $[5.654, 6.374]$ for nominal orders two, four, and six. Direct suppressed derivatives at both signs agree to 2.22×10^{-16}, rotation covariance to 2.22×10^{-16}, and all 43 named axis-limit checks to 1.39×10^{-17}. Raw quotient evaluation at the closest cell-centered diagnostic point, $|\rho| = 0.0125$, has residual 2.04×10^{-12} from expected floating-point cancellation and passes the explicit diagnostic-only 10^{-10} tolerance. A division-free degree-six Taylor reference remains below 2.38×10^{-11} on the complete adjacent- cell sequence. Neither tolerance qualifies a production regularization. These results define an oracle, not production qualification; the integrated provider must reproduce them independently.

2.5 Reproduction amplitudes and claim boundary

The exact arXiv v1 source contains the TeX and vector figure PDFs but no machine-readable numerical series. It explicitly identifies the Figure 1 upper-branch amplitudes $\overline{1.30080779}$ (dispersing) and $\overline{1.300807615}$ (apparent-horizon confirmed), the Figure 3 Brill case $A = -0.047$ with $\rho_0 = 5$, and the Figure 5 highlighted best subcritical case $\overline{1.3008077675}$. The remaining Figure 4 markers and most Figure 5 curves have no exact amplitude labels. They remain anonymous in the provenance table and are not converted from pixel positions into simulation inputs. This is a limitation of exact amplitude recovery, not permission for a new threshold bisection. The cited $\rho_0 = 5$ predecessor supplies classified bounds $A_{\mathrm{sub}} = -0.04878$ and $A_{\mathrm{sup}} = -0.04900$, which support the subcritical placement of $A = -0.047$. Exact same-family values such as $A = -3.50909$ are valid campaign inputs even when their identity with an anonymous Figure 4 marker is unknown. For the current upper branch the classified endpoints $\overline{1.3008077675}$ and $\overline{1.300807615}$ define the derived plotting midpoint 1.30080769125, full width 1.525×10^{-7}, and half-width 7.625×10^{-8}. The midpoint is not a published run amplitude.

3 Frozen Cartesian AthenaK baseline

Before integrating the Cartoon backend, we froze AthenaK commit `594045ef9c051739911d5efcf5b3fe8dd38ffb59` with Kokkos commit `6739bc623081648af9e752b616d9671527922cbf`. The authoritative Perlmutter baseline is Slurm job `56550169`, run on one node with four MPI ranks bound to four distinct 40 GB A100 GPUs. Fresh CUDA and CPU builds were made inside compute steps. The run root is `/pscratch/sd/h/hzhu/collapse-critical-perlmutter/runs/axisymmetric-cartoon-z4c-baseline-56550169`. All 106 entries in its `SHA256SUMS` file verified after completion.

Two earlier attempts are retained as harness evidence rather than numerical results. Job `56549641` stopped before configuration because its build step requested more CPUs per task than the allocation registered. Job `56549724` completed both builds and all second-order cases, then exhausted one A100 before sixth-order evolution when a single rank reserved capacity for 4096 MeshBlocks. The accepted retry divided the same global capacity over four ranks as 1024 MeshBlocks per rank and used all four GPUs for every CUDA case.

The reported cold and warm sixth-order throughputs were respectively 3.169070×10^6 and 3.191818×10^6 zone-cycles per second, a 0.718% increase in the warm repeat. This pair defines the pre-integration workload; it is not by itself a performance comparison with the future provider-refactored Cartesian path.

Table 2: Frozen Cartesian linear-wave errors from job 56550169. The quasi-two-dimensional cases have $n_{x^3} = 4$ and are historical slab regressions, not true two-dimensional Cartoon evolutions.

Case	RMS–L_1 error	refinement ratio
CPU slab, $32^2 \times 4$	2.395272×10^{-10}	—
CPU slab, $64^2 \times 4$	5.838675×10^{-11}	0.2437583
CUDA 3-D, order 2, 32^3	1.424620×10^{-10}	—
CUDA 3-D, order 2, 64^3	3.159703×10^{-11}	0.2217927
CUDA 3-D, order 6, 64^3	5.094343×10^{-12}	—
CUDA 3-D, order 6, 64^3 warm	5.094343×10^{-12}	—

The existing boosted-puncture and FastFlow regression also passed. Its final norms were $C = 1.79735 \times 10^{-2}$, $H = 4.65582 \times 10^{-3}$, $M = 1.37659 \times 10^{-3}$, $Z = 2.97762 \times 10^{-3}$, $M_x = 1.01421 \times 10^{-3}$, $M_y = M_z = 4.43288 \times 10^{-4}$, $\Theta = 3.0628 \times 10^{-5}$, and the final horizon RMS residual was 2.7206351×10^{-2}. Each value satisfies the unchanged upstream threshold recorded in `baseline_verdict.json`.

The external attestation `artifacts/axisymmetric_cartoon_z4c_2026-08-10/baseline/perlmutter_baseline_56550169_attestation.json` binds the job, source and submodule identities, executable and CMake-cache hashes, four GPU UUIDs, numerical values, and run-root checksum digest. These data qualify only the unchanged Cartesian functional baseline. They are not evidence that a Cartoon backend exists or is correct.

4 Production architecture and R1 qualification

The fixed architecture evolves vacuum Z4c on the full signed meridional (ρ, z) plane with a single stored suppressed direction. The axis is an internal cell face, Cartesian 3-D remains the default, and the production path must dispatch symmetry and stencil order on the host rather than branch per cell. Native AthenaK quadtree AMR, task scheduling, communication, restriction, prolongation, load balancing, restart, diagnostics, and output remain the consuming infrastructure.

The pushed R1 source slices fix refinement decisions on the collapsed Cartoon plane, add parity-aware central diagnostics and the signed physical-volume measure, and provide the compact R1 campaign driver. These changes are carried by logical commits `4851799c7b45df0800d47a5b75d01a685b66e160`, `143f71f4f7cf5db10ad8fd55c79eda47635cac2a`, and `1077dd11534153fc37a653224b6054c3d39cbd25`. Focused source and mathematical checks passed. The accepted source-bound CUDA+MPI4 R1 evidence contains all six declared fixed, AMR, restart, and derefine cases plus all three exact equality pairs. Its campaign-state and analysis SHA-256 values are, respectively, `6259ac8c2068f847010c2cbd84b339bc24ffa249943616b0d689df59df1add7e` and `0b798b2029543d5869da4f5cb787b5c465c43bd48fdcb71966f7b7725ab0c2bd`. The accepted analysis binds tree symmetry, rank ownership, central lapse and proper time, constraints, curvature, physical-volume history, and exact fresh/restart state. Earlier failed trees and restart attempts remain in the campaign ledger; they were not deleted to obtain this disposition.

5 Kerr puncture and apparent-horizon qualification

The reusable initial-data target is the stationary, axis-aligned single-hole Kerr puncture construction of Liu, Etienne, and Shapiro [5]. The intended public parameters are mass M, dimensionless spin $\chi = a/M$ with $|\chi| < 1$, axial center z_h, and the two documented initial-gauge choices. The moving-puncture default is $\alpha = \psi^{-2}$ and $\beta^i = 0$. The coordinate transformation uses

$$r_{\mathrm{BL}} = r\left(1 + \frac{r_+}{4r}\right)^2, \qquad r_{\mathrm{horizon}} = \frac{r_+}{4}, \tag{24}$$

without arbitrary clipping of the puncture.

That pgen is now integrated as a reusable, input-selected source component. It implements Eqs. (11) and (13)–(15) of Ref. [5] for one stationary axis-aligned hole, with $M > 0$, $|\chi| < 1$, signed spin, and center z_h. One physical Cartesian evaluator feeds both Cartesian 3-D and the signed-ρ Cartoon component map. The production code uses analytic conformal limits at the puncture and no epsilon clipping; a topology check rejects a cell center exactly on the puncture where the physical ADM fields are unavailable. The default precollapsed gauge is $\alpha = \psi^{-2}$ and $\beta^i = 0$; an optional stationary analytic lapse and shift implements the paper's Eqs. (6) and (7). The pgen implementation is recorded by commits `ae42f3eac2b5944e93fe0726132cb0ca62a81b5c` and `2a311309479fedd74ad3acea7451eeea3807ec8a`.

The R2 source also delegates Cartoon horizon finding to the existing FastFlow scheduler through an $m = 0$, θ-only $a_{\ell 0}$ adapter with analytic 2π azimuthal integration. It uses the shared parity-aware meridional sampler for physical Cartesian ADM and extrinsic-curvature data, evaluates an independent direct expansion residual, and archives area, irreducible and Christodoulou masses, spin S_z, radius, center, and shape. Deterministic origin and positive/negative-axis lapse-extremum candidates are supported. Restart schema 2 persists the selected spectral surface and first-found time, while legacy active carriers fail closed. The implementation, runtime delegation, campaign tooling, and restart corrections are commits `3e7d32ac32b6f602f073b4010684f6133006e9f0`, `e39edc4897023533e9f4e27084f7fc7150affbe6`, `fcde7fbe803469047228048afe5b88fca5bdeaea`, `aa42bafdb4c28ca48d741baa684e3b3bc55a136d`, and `47d88819285f5d733d3777a7f51a781bf271190a`. Focused oracle, limit, spin/mass, finite-value, and restart self-tests passed. The integrated test-only corrections are commits `ca5ede9b0a357380218d21cb29362e773ac914c6` and `138f4d54e20bfb70b9d2691b82b022e4de06d956`. V6 stopped before campaign preparation. Subsequent source-bound evidence accepted R1 and the user authorized R2 as "accepted with an explicit lower-basis waiver." The original all-basis gate remains unpassed: displaced $(4, 16)$ failures are retained, while the required off-axis evidence is the accepted displaced $\pm(6, 24)$ pair and mirror check. The L8 supplement is corroborating characterization only and is not used to erase the L4 failure.

Evidence pending—R2 all-basis limitation. The original all-basis horizon gate is not passed because displaced $(4, 16)$ remains failed evidence. Preserve the explicit lower-basis waiver and do not relabel it as full all-basis qualification. This text is a report placeholder, not a numerical result or qualification claim.

A black-hole case will be accepted only with finite evolution, controlled constraints, and converged horizon mass, spin, radius, center, shape, flow residual, and direct expansion residual. Distributional puncture samples are archived separately rather than treated as ordinary pointwise constraints.

6 Half-plane Kerr default-gauge resolution experiment

A separate pushed half-plane source branch was tested after the main evidence freeze. Its exact commit is `f95a05802621b76cff2894d562c38df4b0d09661`, tree `4a03dc2ec051463372d812cfd053ca4db3d0698a`. This branch uses $\rho \geq 0$, exact parity ghosts, centered finite differences, and analytic SO(2) reconstruction without near-axis fitting. The physical case is a single Liu–Etienne–Shapiro Kerr puncture with $M = 1$ and dimensionless spin 0.5. The initial lapse is pre-collapsed, $\alpha = \psi^{-2}$, and the initial shift is zero.

At the user's request, this experiment uses AthenaK's default direct moving-puncture gauge rather than the paper-matched, telegrapher, or scale-selective alternatives:

$$\partial_t \alpha = \beta^i \partial_i \alpha - 2\alpha \widehat{K}, \tag{25}$$

$$\partial_t \beta^i = \widetilde{\Gamma}^i + \beta^j \partial_j \beta^i - 2\beta^i. \tag{26}$$

The exact switches are `lapse_oplog=2`, `lapse_harmonicf=1`, `lapse_advect=1`, `shift_Gamma=1`, `shift_advect=1`, and `shift_eta=2`. Telegraph and slow-start lapse, the auxiliary B^i driver, scale-selective damping, and chi flooring are disabled. The AMR trigger is `dchi_max=0.02`, not 0.05. The scheme is O6 with RK4 and a target time of $5M$.

The prospective resolution sequence was finest spacing $M/32$, $M/48$, and $M/64$. The first case reached $5M$. The $M/48$ case failed at cycle 2178, $t \simeq 4.560417M$, when the central diagnostic stencil became nonfinite. The $M/64$ case was not run because the controlling contract required a stop once convergence had already failed. Table 3 records the resulting claim boundary.

Table 3: Default-gauge half-plane Kerr resolution status. The horizon values are the first accepted origin candidate, not a late-time qualification result.

Grid	End time	Status	$M_{\rm AH}/M$	$S_z/M_{\rm AH}^2$
$M/32$	5.0000	completed; late constraints large	1.001947	0.509555
$M/48$	4.5604	failed central finite gate	1.000492	0.501070
$M/64$	—	not run after failed convergence gate	—	—

Figure 1 shows the native AthenaK history diagnostics. In the left panel, solid, dashed, dotted, and dash-dotted curves are the absolute reported C, H, M, and Z `norm2` columns, respectively. The finer run initially has smaller errors but develops a terminal spike. The right panel separates the reported axis and off-axis C RMS diagnostics; the growth is concentrated on the axis side rather than being an outer-boundary arrival.

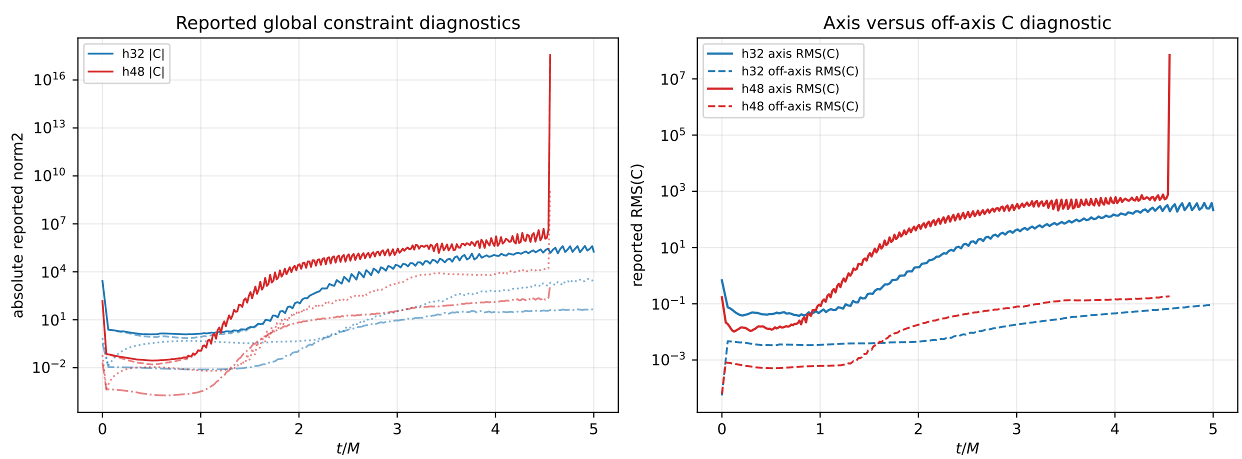

Figure 1: Constraint histories for the existing $M/32$ and $M/48$ default-gauge runs. These are reported AthenaK history diagnostics; no post-processing floor or masking is applied.

Apparent horizons are found early and have the expected mass and spin, with the $M/48$ values closer to $(1, 0.5)$ than the $M/32$ values. This is useful characterization but not horizon convergence: the origin finder last accepts at $1.95625M$ on $M/48$ and at $2.834375M$ on $M/32$. The final origin attempts have direct residuals 0.07895 and 0.07975, respectively, and fail the iteration limit. Figure 2 therefore plots only accepted mass/spin points and separately retains all residual attempts.

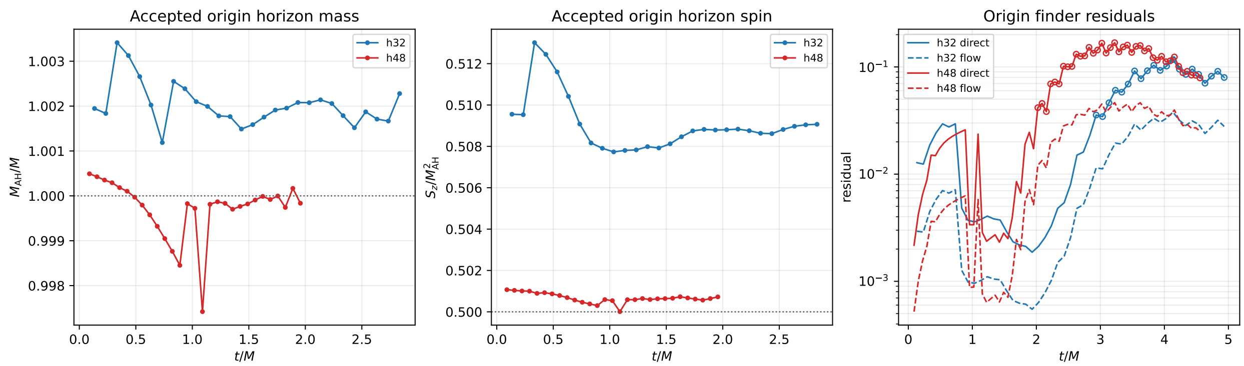

Figure 2: Origin-candidate apparent-horizon mass, spin, and finder residuals. Open residual markers denote rejected origin attempts.

A bounded MPI4 CPU restart from authenticated $M/48$ cycle 2080 reproduces the production CUDA failure at the same cycle and time. Native AthenaK binary outputs every cycle then locate the unstable mode at $\rho = 2.5h = 0.0520833M$ and $z \simeq \pm 0.28M$, inside the apparent horizon. Figure 3 shows the actual post-evolution AthenaK constraint and conformal-extrinsic-curvature cells; it is not an IrisK solver slice or an externally interpolated field. The paired hot spots preserve the equatorial mirror symmetry through the nonlinear growth.

The cycle-resolved growth in Fig. 4 is not caused by reflection asymmetry. Stored-component mirror parity is approximately 10^{-6} or better before the terminal event. The algebraic determinant and trace-free constraints likewise remain small until cycle 2172, when the physical constraint and conformal-A mode has already entered rapid growth.

Default-off, read-only RHS instrumentation was used only in the local replay. Figure 5 shows that the first large $\widetilde{A}_{ij}$ contribution is the trace-Ricci term, reaching 7.33×10^5 at cycle 2170 RK stage 2. The $\widetilde{\Gamma}^i$ equation is dominated by the shift-derivative contraction and expansion terms, reaching 1.40×10^6 and 9.90×10^5. The explicit Gamma damping contribution peaks near 1.06×10^2, orders of magnitude smaller. The immediate trigger is therefore a coupled Gamma-driver/Lie–Ricci puncture-interior RK-stage mode, not demonstrated Z4c damping stiffness, GPU nondeterminism, MPI transfer, outer-boundary arrival, or equatorial asymmetry.

This still does not identify a scientifically justified formula repair. The remaining ambiguity is between a continuum puncture-interior instability of this direct moving-puncture gauge and a nonlinear SO(2) derivative stability defect. No gauge, damping, CFL, dissipation, floor, threshold, or order was tuned after observing the failure. The resolution experiment is therefore *not qualified*: it supplies horizon characterization and an exact failure diagnosis, but no three-grid convergence result.

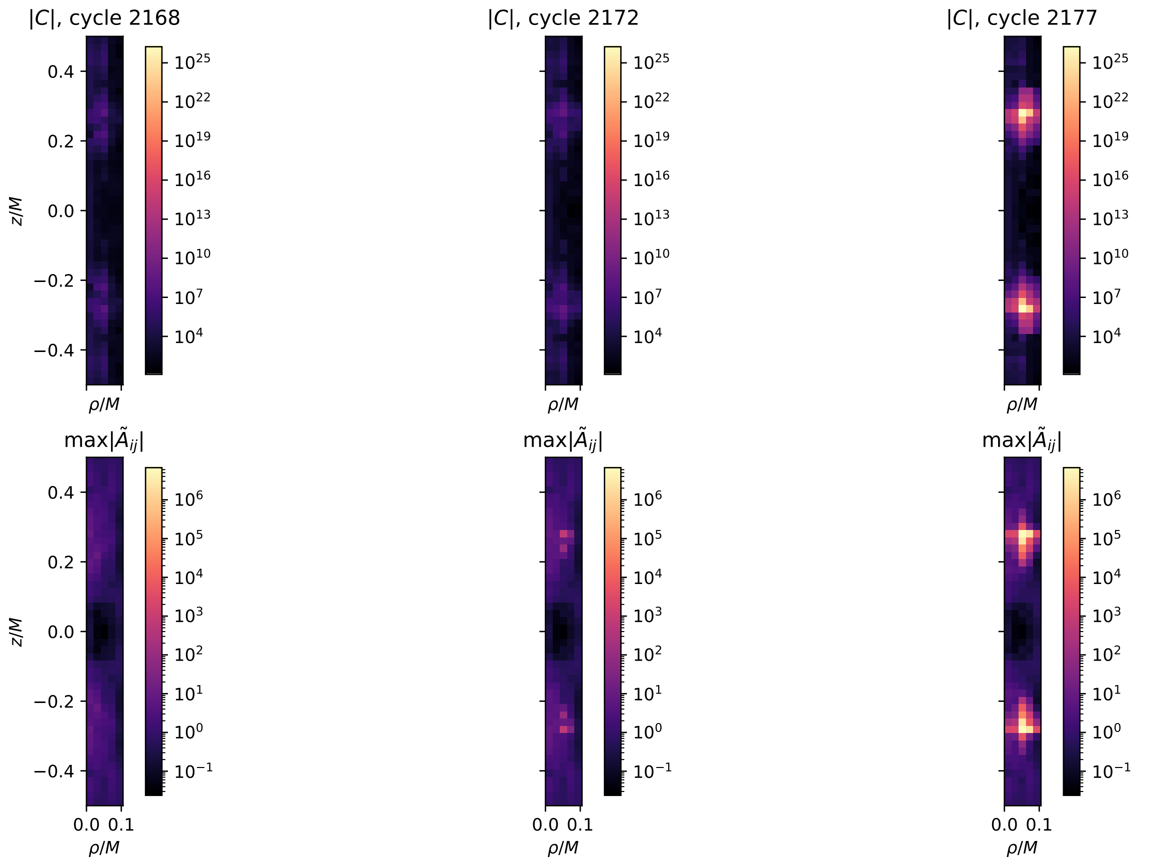

Figure 3: Near-axis native AthenaK cells at cycles 2168, 2172, and 2177. The panels cover the five innermost half-plane radial cells and $|z| \leq 0.5M$.

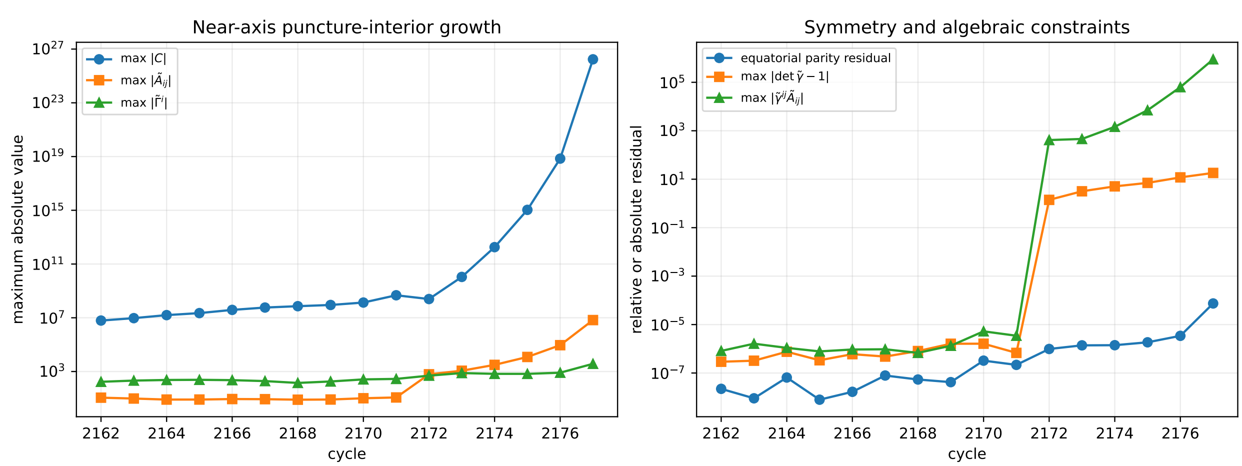

Figure 4: Puncture-interior growth, mirror parity, and algebraic constraints in the bounded CPU replay.

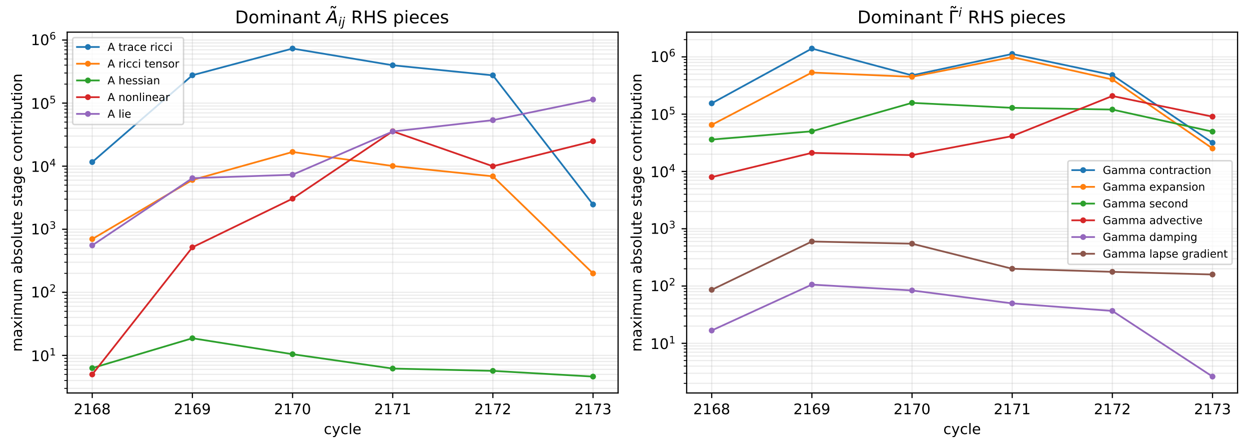

Figure 5: Maximum termwise RHS contributions over all four RK stages and MPI ranks in cycles 2168–2173.

7 Collapse-wave qualification and paper analogues

The official arXiv:2607.10843v1 source tarball and PDF are frozen byte-for-byte at SHA-256 `191d17a04647de81d388a5f29c93f06a2a3cdfbbffc0ebaa4045cee6ebea59ff` and `8fad9241eee63d919eef19fd78a65b43f76e9a6112fbf56d0d3aa8cac0c20517`, respectively. The archive contains its TeX and eleven vector figure PDFs but no machine-readable numerical series [1]. Therefore published caption/table amplitudes and explicitly identified values from cited sources are valid inputs, while anonymous markers may not be reverse-engineered from pixels. The frozen inputs on the critical path are summarized below.

Table 4: Exact literature inputs available before any AthenaK paper run.

Figure use	Family	Amplitude	Literature classification
Figure 3	off-center Brill, $\rho_0 = 5$	-0.047	subcritical bound
Figure 1	time-asymmetric upper branch	1.30080779	dispersing
Figure 1/5	time-asymmetric upper branch	1.300807615	AH-confirmed
Figure 5	time-asymmetric upper branch	1.3008077675	best subcritical
Figure 4	centered negative Brill	-3.50909	exact same-family input
Brill bracket	off-center, $\rho_0 = 5$	$-0.04878, -0.04900$	sub/super

The derived midpoint 1.30080769125 is a plotting reference, not a published run amplitude. Its bracket width is 1.525×10^{-7} and its half-width is 7.625×10^{-8}.

The accepted IrisK Figure-3 Brill import candidate is the 48×32 payload for $A = -0.047$, $\rho_0 = 5$, $z_0 = 0$, and unit widths, from IrisK commit `2a069fd0497ef4352d4ecd28c6879ac47b84a5a1`, tree `87d828305160ff1ae1caca64f6e4c3f4856ed492`. Its payload SHA-256 is `b001be217b9574cde90fe81275541e1ee28601652c18d72a17a346d6405f12c6`; the frozen evidence reports ADM mass 2.660301967997158, collocation Hamiltonian residual $6.9517373601955137 \times 10^{-13}$, and independent fixed-location off-grid residual 0.011179221474641112. This disposition accepts source-faithful initial data for import only; it is not an AthenaK evolution or Figure-3 reproduction.

The repaired importer was then exercised by the source-bound AthenaK/IrisK CUDA+MPI4 v6 campaign at commit `9448c1e62a8335bf7a628d89648c931c0b44ff8a`. Job 56715442 passed the finite import gate and evolved N128 to cycle 544, where the strict positive-chi AMR gate rejected 261,202 parent groups. A local Serial MPI4 replay from the authenticated cycle-512 restart reproduced the exact CUDA partition: 13,216 high-order, 14 limited, 261,202 invalid-parent, and zero invalid-limited groups. At cycle 512 all 304 active blocks were finite and strictly positive in chi, with minimum 0.3640185296535492. A diagnostic census placed before any cycle-544 AMR copy, load-balance transfer, or prolongation found invalid active chi in all 268 blocks selected for refinement and in 209,946 of 274,432 active cells. The failure is therefore a late evolved-state instability, not an AMR counter, GPU, restart, ghost-corner, transfer, or prolongation defect. No floor, clipping, limiter, threshold, or production repair was introduced.

N192 and N256 did not run, and neither the analyzer nor the published-curve comparison ran. Figure 3 is unresolved and has no accepted AthenaK curve. The strict limitation JSON has SHA-256 `d60debc450dcad509a3818d928dbc7b6ac700679c619fee97a342c5fc6927f7f`; its evidence manifest has SHA-256 `8db36d9d63e19a352bb497efb40fe5b09f3e2764238168d4a3d16874880ba0a0`.

The former time-asymmetric source-availability blocker is resolved. The authoritative Khirnov V2 reference source is frozen at Git revision `e631627f70695bde798f22c4baaa472c100e45ef`, tree `3a971d478dc5e5633c0a7cd0307a5b340a56ea8a`, under GNU GPL version 3 or later. The reference library and IrisK converter were built and used to exhaust the prospectively declared candidate pool 24×12, 28×14, 32×16, 36×18, and 40×20 for both Figure-1 amplitudes. All ten field-validation records passed, and the retained coefficient tails decrease, but the promotion gate fails: the relative ADM changes from 36×18 to 40×20 are 0.0105038924902419 and 0.0105039091626585, each greater than the fixed 0.001 (0.1 percent) limit. The authoritative candidate pool is therefore exhausted and blocked. No payload is an accepted import candidate, the gate was not weakened, and no AthenaK import or evolution was run. The complete blocked evidence bundle has detached-manifest SHA-256 `f4f50b804054c8a9a2e4a33a72aea842e168230039b2c4e5930b09a5263a6a4b`. No physical-fate conclusion is inferred from generation labels.

Evidence pending—R3. Resolve the prospectively failed time-asymmetric V2 ADM-stability blocker without weakening its gate, then insert an accepted source-bound payload and handoff manifest, far-subcritical and far-supercritical 2-D runs, the two matched 3-D support runs, and the three-resolution near-critical diagnostic. Process failure must remain distinct from collapse or dispersion. This text is a report placeholder, not a numerical result or qualification claim.

Evidence pending—R4 Figure 1. Insert invariantly matched initial, strong-field, decay, and first-horizon slices for amplitudes 1.30080779 and 1.300807615, including AMR and accepted horizon sections. This text is a report placeholder, not a numerical result or qualification claim.

Evidence pending—R4 Figure 3. Unresolved after the accepted import: the N128 run developed invalid active chi before cycle-544 AMR, and N192/N256 were not run. No AthenaK curve or bamps/prague/sphGR difference is accepted. Further work requires a separately authorized scientific stability resolution, not a weakened positivity gate. This text is a report placeholder, not a numerical result or qualification claim.

Evidence pending—R4 Figure 4. Insert scaling analyses for the centered negative-Brill and upper time-asymmetric families using exact amplitudes only. Fit γ, Δ, or periodic residuals only if the recovered sample supports the fit. This text is a report placeholder, not a numerical result or qualification claim.

Evidence pending—R4 Figure 5. Insert maximum absolute Kretschmann against coordinate and central proper time for every recoverable exact amplitude, marking only accepted horizons. This text is a report placeholder, not a numerical result or qualification claim.

Every future panel must be labeled "AthenaK current-gauge analogue." If a supercritical input has no accepted horizon, its result must be labeled "collapse-like, horizon unresolved" rather than successfully reproduced.

8 Regression and performance

Section 3 records the immutable pre-integration Cartesian baseline. It is not a post-integration performance result.

R5 v10, job 56726255, is retained as a failed predecessor rather than a performance result. Its three current-source linear functional cases passed, and its five baseline samples had median 3,168,116 zone-cycles/s with robust coefficient of variation 0.002097. The boosted functional case then failed the strict-positive chi AMR parent-stencil gate at cycle 1, before any candidate warm-up or timing sample. The immutable limitation JSON has SHA-256 `ff9c790d8c0fa6820349242380abd15aa43a8b39ee7b99a9d71853c8bca393ed`; its root and detached manifest SHA-256 values are `30968c57f5dc4b9b2b519dcbc1e38f8f4f18cac3e115b3290ee961c6cec3fb08` and `80e5a468716a776a6ed2bf8a9d739e0064e5a02188c4f15cf3bce8d3eebaf427`. No performance verdict follows from this predecessor.

The diagnosed boundary-chi repair is integrated at `e8a0253827dba262f5df3157ffc820b4e695127d`. It preserves the existing whole-sibling minmod fallback and fail-closed parent inventory without floors, clipping, or gate changes. Focused Serial and MPI builds and the analytic boosted/Schwarzschild fixtures passed before integration.

V18, job 56741854, then found a CUDA out-of-bounds access in the re-enabled same-level coarse refresh. The final source at `5760da5893b256d32da5fb13f92ed8e0c102ed39` restricts only coarse targets backed by complete stored fine pairs and uses the existing oriented one-sided fourth-order weights at legitimate outer pairs. V19, job 56748297, verifies that the illegal address is gone. Its initial boundary inventory is 194,096 high-order groups, 16 positive limited groups, and zero invalid parents; the first post-RK inventory is 193,912 high-order groups and 200 invalid parents. It fails closed before candidate timing. Aggregate evidence cannot resolve whether the parents contain invalid updated chi or an invalid coarse restriction representation, so no further safe single repair is claimed and the result is not called a physical instability.

The v19 five-sample baseline is 3,157,141, 3,176,336, 3,171,336, 3,188,219, and 3,158,748 zone-cycles/s. Its median is 3,171,336, MAD is 12,588, and robust coefficient of variation is 0.00588489, below the prospective 0.02 dispersion gate. This is baseline characterization only: candidate warm-ups and samples are zero, and the candidate-to-baseline ratio, matched 3-D/2-D cost ratio, and performance verdict are all null. The terminal limitation JSON has SHA-256 `b27c55c70030de6ffdc39e852e15dab0bd8641bb5262054bd2e24ebd013145c1`. R5 is not accepted.

The release analysis will retain ordinary runtime, MeshBlock history, throughput, resource, and process evidence. It will not introduce HBM monitoring or rerun accepted simulations merely to change plotting logic.

9 Provenance and reproduction

The report began from the following accepted identities:

AthenaK commit	`0d21e193bf2065427e0975c973df875ae33c7db3`
AthenaK tree	`ab1868ed34632aa54eb79f9f0fd4760507acdf02`
Kokkos commit	`6739bc623081648af9e752b616d9671527922cbf`

These are starting identities. The historical importer/R1/R2 integration freeze was:

AthenaK branch	`codex/axisym-cartoon-integration-20260810`
AthenaK commit	`9448c1e62a8335bf7a628d89648c931c0b44ff8a`
AthenaK tree	`36e6dfc300e771a2093d6b1bd98da631a70554d7`

This is a historical import freeze. The current authoritative identity is:

AthenaK branch	`codex/axisym-cartoon-integration-20260810`
AthenaK commit	`5760da5893b256d32da5fb13f92ed8e0c102ed39`
AthenaK tree	`9318e083497997edb80d93c913934acd987699eb`
Kokkos commit	`6739bc623081648af9e752b616d9671527922cbf`

This is the verified pushed final evidence identity. It is not an R5-qualified release identity because the terminal v19 gate did not pass.

The later half-plane Kerr resolution experiment is a distinct pushed branch, not a relabeling of the authoritative R5 source above:

AthenaK branch	`codex/cartoon-half-plane-so2-20260813`
AthenaK commit	`f95a05802621b76cff2894d562c38df4b0d09661`
AthenaK tree	`4a03dc2ec051463372d812cfd053ca4db3d0698a`
CUDA executable SHA-256	`c14f6ac88144ad27b097e6f95249167a37d32cc50de2923894e66f10de2adeba`
Perlmutter job	56882670

Its $M/32$ and $M/48$ inputs have SHA-256 values `769843e2ebc584b1d4294e8b1ddbf9f1d75f8962b8ece647fdf8176c7d06d0c5` and `f026ff8e26d4ed4ece9c8e7c183d07c16145c7dafa3c57c2ae0eb4e631f41671`. The campaign-state and local diagnostic-summary SHA-256 values are `ed35bcedc3f107cb2e34bdeaf2764ce7d82a203d6bb6cccf4426eb58dc62ac6a` and `1c5f7b02182be4d77ae091024ac59376f7b4e64850740c0d1ed66ac7dc9ab928`. The latter binds all parsed history, horizon, native slice, and RHS-term input bytes. It records a failed convergence qualification and leaves the $M/64$ case unrun.

Table 5: Logical R1/R2 integration slices at the source freeze.

Commit	Subject
`4851799c7b45df0800d47a5b75d01a685b66e160`	Fix Cartoon AMR on collapsed plane
`ae42f3eac2b5944e93fe0726132cb0ca62a81b5c`	Add reusable Kerr puncture initial data
`2a311309479fedd74ad3acea7451eeea3807ec8a`	Enable reusable Kerr puncture pgen for Cartoon
`143f71f4f7cf5db10ad8fd55c79eda47635cac2a`	Add Cartoon central diagnostics and volume measure
`1077dd11534153fc37a653224b6054c3d39cbd25`	Add Cartoon R1 campaign tooling
`3e7d32ac32b6f602f073b4010684f6133006e9f0`	Add Cartoon $m = 0$ FastFlow adapter

Commit	Subject
e39edc4897023533e9f4e27084f7fc7150affbe6	Integrate Cartoon $m = 0$ FastFlow runtime
fcde7fbe803469047228048afe5b88fca5bdeaea	Add Cartoon R2 horizon campaign tooling
aa42bafdb4c28ca48d741baa684e3b3bc55a136d	Update R2 campaign for FastFlow restart schema 2
47d88819285f5d733d3777a7f51a781bf271190a	Preserve FastFlow restart override conflicts
cccbeadb57d79af086794e9e8f016caa939e64de	Fix Cartoon MMS CUDA tensor mapping
01a03005c5abcac3848f6e52d8cf9416a2483186	Enable input-selected Iris ADM import
1cc610ba2350da657d85fefbdd0c482c548b0068	Fix Cartoon FastFlow CUDA lambda access
6fa8503d8ce95c56636534b18a77256bd56ba0bd	Map Iris ADM data into Cartoon coordinates
f5b4b377c32f5a2c18bf855f409aa18361896534	Add Brill Figure-3 Cartoon campaign tooling
194b6e1f087570017d68d310554d39091be08955	Resolve relocated Brill provenance safely
ca5ede9b0a357380218d21cb29362e773ac914c6	Preserve preallocation failure-order fixture
138f4d54e20bfb70b9d2691b82b022e4de06d956	Fix legacy FastFlow restart fixture mutation
a008352328a3cbac01fbcaf2a6f8020fcc3b58ab	Preserve historical horizon find time on restart
9448c1e62a8335bf7a628d89648c931c0b44ff8a	Bound collapsed Iris imports to stored cells
3c597728909192dced0c087b677f6a4e3f5c0a9c	Fix R5 Cartesian MeshBlock capacity
e8a0253827dba262f5df3157ffc820b4e695127d	Preserve positive Z4c chi at coarse-fine boundaries
66094dc2b46a42f96c88e9a4b24819109647bb81	Refresh Z4c same-level coarse corners
5760da5893b256d32da5fb13f92ed8e0c102ed39	Bound same-level coarse restriction to stored pairs

The accepted Brill handoff source is IrisK commit `2a069fd0497ef4352d4ecd28c6879ac47b84a5a1`, tree `87d828305160ff1ae1caca64f6e4c3f4856ed492`. Its import-only disposition and the separately frozen Khirnov V2 source are recorded in Section 7.

The retained Perlmutter sequence distinguishes production compilation from the test harness and from numerical qualification:

- Job 56666406 (v5), source `01a03005c5abcac3848f6e52d8cf9416a2483186`, configured and then failed in the FastFlow CUDA compilation step. Its terminal accounting prefix is two numbered steps, no executable or focused test exists, and no case ran. The remote root and detached manifest

SHA-256 values are `b701a40937be4bd2bfaae7adea2302993f8f7dc8fd8239a05bc012a2e248e6a0` and `966ad5976c5dc19a2190a129d36280c53fe2d910eba8cc530f9c527fdce34c6e`.

- Job 56668445 (v6), source `f5b4b377c32f5a2c18bf855f409aa18361896534`, completed configuration, the full CUDA build, and the CUDA+MPI executable backend check. The next step ran the seven focused tests and failed four: three CMake-generated Python commands used `/usr/bin/python3` version 3.6, while one test retained the superseded preallocation diagnostic ordering. Campaign preparation never began and no R1/R2 case ran. The root and detached manifest SHA-256 values are `3d73c9bf659cac1cacc7d64dd1fd83a4c296df249c28208d97b99dc61662b46c` and `891231ced4c1fd60986ca20c52fd2f9ad046bd0a16720da68f0b2f84a5c16de0`.

- Later settled CUDA+MPI4 campaigns accepted all six R1 cases and their three exact fresh/restart equality pairs. The composite R2 disposition is accepted under the explicit lower-basis waiver; it preserves the L4 failure, uses displaced L6 positive/negative and mirror evidence for closure, and retains L8 as corroboration only.

- R5 v10, job 56726255, is an immutable failed predecessor. Its limitation JSON, root manifest, and detached manifest SHA-256 values are `ff9c790d8c0fa6820349242380abd15aa43a8b39ee7b99a9d71853c8bca393ed`, `30968c57f5dc4b9b2b519dcbc1e38f8f4f18cac3e115b3290ee961c6cec3fb08`, and `80e5a468716a776a6ed2bf8a9d739e0064e5a02188c4f15cf3bce8d3eebaf427`. Candidate timing did not start. The diagnosed chi-boundary repair is integrated at the authoritative source above.

- R5 v18, job 56741854, retained six completed baseline steps but stopped in boosted initialization with a CUDA illegal address. Source tracing identified an out-of-bounds same-level coarse restriction. Its root and detached manifest SHA-256 values are `9e5a148bcb730f720f9e6b5de70eb15931c51958f12505661c34ecc66cd94b4c` and `609eb4bd4945f9e92a191c9a4e97b258592acdd95647d8db7e8184803faaf5ae`.

- Final R5 v19, job 56748297, used source `5760da5893b256d32da5fb13f92ed8e0c102ed39`, executable SHA-256 `8cb218d37a908c85377fdc63651b8bcf3ad489b7cc816e37d17b681a7da32130`, and unchanged boosted input SHA-256 `8e7da544c3e5f6cecb4312a4a21819bf6c77bdb3ed6372f81ac2ce72d1ec05a8`. Configuration, build, focused tests, and six baseline steps completed. Step 6 then failed the first post-RK boosted boundary pass on 200 invalid parent stencils. The root, detached, settled-accounting, campaign-state, and run-log SHA-256 values are `dc6e8d620e7c203bdf4f19d4f482860282cb865b1896429cc92ec450f523c049`, `c6d2f6b9f41e4bd1f96e79d7da7378ab7dc8ee2a10b4ad1408befee8d6da27c1`, `b2afd89a53111453479b3d5d9af8c266af50a58c4ebd88b970088977addc0f8a`, `ee503df98d1f5c7f12c610f86f9ece8de70f6cbfa1c9253a9c7f24cf8c11d9d6`, and `5bb56bd2082ab24fe2efd82c3c744eb1f1400850a6f5bf8ce451e7d837489c88`. No candidate timing or performance verdict exists.

The earlier job 56661702 remains archived compile-failure evidence for the MMS tensor-map repair, but it is not a runtime qualification result.

The controlling arXiv:2607.10843v1 PDF and source-archive SHA-256 values are, respectively:

`8fad9241eee63d919eef19fd78a65b43f76e9a6112fbf56d0d3aa8cac0c20517`

`191d17a04647de81d388a5f29c93f06a2a3cdfbbffc0ebaa4045cee6ebea59ff`

The accepted derivative report is preserved at SHA-256:

`03ee49019a7cd6ed7498e21ba6d98e4ffb2d6e96a44ef6f6df1ec50955946b5f`.

The machine-readable claim map is `artifact_manifest.json`. Pending claims and data products are listed separately in `placeholders/claims.csv` and `placeholders/data.csv`; a placeholder may be promoted only after its immutable evidence path and checksum are known.

The terminal R5 limitation is frozen under `../../artifacts/axisymmetric_cartoon_z4c_2026-08-10/continuation/r5_cartesian_v19_limitation_2026-08-12`. Its JSON, manifest, and detached-manifest SHA-256 values are `b27c55c70030de6ffdc39e852e15dab0bd8641bb5262054bd2e24ebd013145c1`, `2b6a84607e1a8a4ddde92b37161823ac4677d943fd4680eb8aa4ba00e69a1ec6`, and `d9a913642c169e8be49e13f6246bfbb090c4c3388d2e5feee7ee618e7dc282d2`.

The deterministic local build entrypoint is `build_report.sh`; exact use and the literature checksum audit are documented in `README.md`.

10 Conclusions and remaining work

At this freeze, the literature, exact-amplitude table, analytic SO(2) oracle, accepted derivative report, historical Cartesian baseline, and pushed R1/R2 source implementation are reusable. The source includes the reusable Kerr puncture pgen, Cartoon AMR/diagnostics tooling, and the $m = 0$ FastFlow/restart path at pushed commit `5760da5893b256d32da5fb13f92ed8e0c102ed39`, tree `9318e083497997edb80d93c913934acd987699eb`. V5 exposed a FastFlow CUDA compile defect; v6 demonstrated a complete CUDA build and backend check before stopping in the test harness with no cases; the narrow test repairs are integrated. Later campaign evidence accepted R1 and accepted R2 with the explicit user-authorized lower-basis waiver; the original all-basis R2 gate remains unpassed and the displaced $(4, 16)$ failures remain visible. Milestone 2 remains explicitly waived and unpassed.

The Brill Figure-3 importer passed its CUDA runtime gate, but the N128 evolution failed at cycle 544 after active chi became invalid before AMR. Serial MPI4 reproduced the exact failure and a pre-AMR census excluded AMR transfer, prolongation, counter, GPU, restart, and ghost-corner defects. N192/N256 and the curve analyzer did not run, so Figure 3 is unresolved with no accepted curve. The authoritative time-asymmetric V2 library build and bounded generation succeeded, but its prospective pool is exhausted without an accepted import candidate: both 36x18 to 40x20 ADM changes exceed the fixed 0.1 percent stability gate. That milestone is blocked, not pending a routine launch, and the threshold was not weakened. R3 collapse-wave execution, the R4 Figure 3 analogue and Figures 1/4/5 therefore remain incomplete or blocked. R5 v10 exposed a coarse/fine chi-boundary positivity seam and v18 exposed a CUDA out-of-bounds access in the subsequent same-level refresh. V19 verifies the bounds repair but fails closed after the first RK stage with 200 invalid chi parent stencils. No further exact copy/order seam is proven, no physical-instability conclusion is made, and no floor, clipping, formula, gate, or threshold was changed. Candidate timing did not start; therefore R5 is not accepted and no performance or release verdict is claimed. The final report and reproducibility bundle preserve this terminal limitation rather than converting incomplete evidence into a pass.

The later half-plane Kerr resolution experiment is likewise retained as a diagnosed nonqualification. It used the requested default advective 1+log lapse and direct Gamma-driver shift with pre-collapsed initial lapse and `dchi_max=0.02`. Although $M/32$ reached $5M$ and early $M/48$ horizon mass/spin were closer to the analytic targets, $M/48$ failed at $4.5604M$. A CPU restart reproduced the failure and cycle/stage diagnostics localized a symmetric puncture-interior Gamma/A–Ricci mode. Explicit Gamma damping was orders of magnitude smaller than the dominant shift-derivative and trace-Ricci terms. Because the evidence does not distinguish a continuum puncture-interior gauge mode from a nonlinear SO(2) derivative stability defect, no safe formula repair is claimed, $M/64$ was not run, and no three-grid constraint or horizon convergence result exists.

A Evidence placeholder index

This appendix is intentionally nonnumerical. It keeps the final report shape stable while preventing prose from getting ahead of evidence.

ID	Gate	Promotion requirement
R1-001	AMR/restart	Accepted source-bound campaign evidence
R1-002	diagnostics	Accepted source-bound campaign evidence
R2-001	Kerr pgen	Accepted with explicit lower-basis waiver; all-basis gate not passed
R2-002	horizons	Displaced $(4, 16)$ failures retained; $\pm(6, 24)$ accepted
KERR-HP-001	half-plane convergence	$M/48$ failure retained; no $M/64$ or convergence claim
R3-001	collapse	Figure 3 unresolved after N128 late-state failure; V2 pool exhausted at failed ADM-stability gate
R4-001	Figures 1/3/4/5	Machine-readable plot data and difference metrics
R5-001	regression	Terminal v19 limitation; no candidate performance result
R5-002	release	Final limitation manifests, hashes, and deterministic PDF audit

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